

Laboratory work - Spring 2026

The main goal of laboratory work is to present the most important aspects of data science in practice and to teach you how to use key tools for a NLP engineer. We especially emphasize on self-paced work, raising standards related to development, replicability, reporting, research, visualizing, etc. Our goal is not to provide exact instructions or "make robots" out of participants of this course. Participants will need to try to navigate themselves among data, identify promising leads and extract as much information as possible from the data to present to the others (colleagues, instructors, companies or their superiors).

Important links

[Lab sessions course repository](#) (continuously updated, use weekly plan links for latest materials)

[Project report template](#)

Books and other materials

[Speech and language processing](#) (online draft)

[Python 3 Text Processing with NLTK 3 Cookbook](#)

[Introduction to Data Science Handbook](#)

[Razvoj slovenščine v digitalnem okolju](#) (February 2023)

Previous years NLP course materials

[NLP course 2024 project reports](#)

[Marks \(public\)](#)

Peer review

TODO: Link to the form (Form closes **Monday, June 1, 23:59**)

** Check Marks spreadsheet (sheet 'Repositories to review') to get list of repositories you need to review.*

Weekly plan

This plan is regularly updated.

Lab sessions are meant to discuss materials and your project ideas. Those proficient in some of the topics covered during the course are expected to help other students during the lab work or in online discussions. Such contributions will also be taken into account. During the lab sessions we will show some DEMOs based on which you will work on your projects. Based on your proposals / new ideas we can adapt the weekly plan and prepare additional materials.

All the lab session tutorials will be regularly updated in the [Github repository](#). During the lab sessions we will briefly present each week's topic and then mostly discuss your project ideas and work. You are expected to check/run notebooks before the lab sessions and then ask questions/discuss during the lab sessions. In the repository's README you can also find the recordings of each topic.

Week	Repository notebooks and materials	Assignments
16.2. - 20.2.	/	
23.2. - 27.2.	Lab work introduction	Assignment 1: Text processing

	Projects overview Group work and projects application procedure Basic text processing Slovene text processing	Assignment 1: Solutions
2.3. - 6.3.	Text clustering Text classification	Projects sign up form (deadline Friday midnight). Github classroom assignment (deadline Friday midnight , only one group member creates a team, exactly three members for a group!). Use GROUP ID from the projects sign up form as a team name. Assignment 2: Text representation Assignment 2: Solutions
9.3. - 13.3.	Word embeddings NNs for sequence modeling (RNNs, GRUs, LSTMs) Multiple simple NN architectures example (Google Colab)	Assignment 3: Word embeddings Assignment 3: Solutions
16.3. - 20.3.	Introduction to PyTorch PyTorch Lightning SLING tutorial (setup, Singularity, SLURM) First submission (Friday, 23:59)	Assignment 4: Attention Assignment 4: Solutions
23.3. - 27.3.	First submission defense (in person)	
30.3. - 3.4.	Introduction to Transformers (tokenization, fine-tuning BERT) Basics of text generation Text generation examples	Assignment 5: Encoding Assignment 5: Solutions
6.4. - 10.4. (Mon. holiday)	Quantization & Basic prompting strategies Retrieval Augmented Generation (RAG) Efficient inference engine (vLLM)	
13.4. - 17.4.	Efficiently Fine-Tuning Large Language Models (PEFT) Introduction to Agents	No lab session on Thursday 12pm. Please attend other sessions.
20.4. - 24.4.	Graph neural networks for text processing	No lab session on Tuesday 12pm. Please attend other sessions.
27.4. - 1.5. (Mon. & Fri. holiday)	NVIDIA Selected part of the Teaching Kits Consultations/Project work/Discussions TBA: 2 consultation slots (Please attend the lab sessions and discuss your work and ideas!) Second submission (Friday, 23:59)	
4.5. - 8.5.	Second submission defense (in person)	

11.5. - 15.5.	<i>Consultations/Project work/Discussions</i> <i>(Please attend the lab sessions and discuss your work and ideas!)</i> Lab session ONLY on Thursday 10am.
18.5. - 22.5.	<i>Consultations/Project work/Discussions</i> <i>(Please attend the lab sessions and discuss your work and ideas!)</i> No lab session on Tuesday 12pm. Please attend other sessions. Final submission deadline (Friday, 23:59) IMPORTANT: Put your repositories visibility to public before the deadline or shortly after! Peer review submission deadline (Monday, May 25, 23:59) Peer review link (see above)!
27.5.	No lab sessions Final project presentations (during lectures slot, P22 at 11:00)

Course obligations

Please regularly check *Weekly plan* and course announcements for possible changes. You are expected to attend the sessions but you **must** attend the defense sessions. At the assignment defense dates at least one member of a group must be present, otherwise all need to provide their doctor's justification. At the last assignment all members must be present and also need to understand all parts of the submitted solution.

All the work must be submitted using your Github project repository. Submission deadlines are indicated in the table above. Submission defenses will be held during the lab sessions.

Students must work in groups of **three members**! There can exist only one group of two members per project type. The distribution of work between members should be seen by commits within the repository.

<i>Obligation</i>	<i>Description</i>	<i>Final grade relevance</i>
Submission 1	Project selection & simple corpus analysis - Group (three members) selection - Report containing Introduction, existing solutions/related work and initial ideas - Proposed project dataset/corpus/data/... - Well organized repository	<i>Pass/Fail</i>
Submission 2	Initial implementation / baseline with results	<i>Pass/Fail</i>

	- Updated Submission 1 parts - Implemented at least one solution with analysis - Future directions and ideas - Well organized repository	
Submission 3	Final solution and report	80%
	- Final report incl. analyses and discussions - Fully reproducible repository	
Peer review	Evaluate your peer group's work	20%
	- Each group will check final submissions of two other peer groups having the same topic	

Total: 100%

Grading criteria

All the graded work is group work. All the work is graded following the scoring schema below. To successfully pass the laboratory work, you need to pass Submission 1 and 2, and gain 50% total from **each of** Submission 3 and Peer review.

Use PUBLIC GROUP ID for public communication regarding your group. GROUP ID is your internal id of a group for which marks will be publicly available.

Scoring

Scoring is done relative to achievements of all the participants in the course. Instructions will define the criteria the participants need to address and exactly fulfilling the instructions will result in score 8. All the other scores will be relative to the quality of the submitted work. The role of instructors is to find an appropriate clustering of all the works into 6 clusters. To better illustrate the scoring, schema will be as follows:

- 10 - exceptional: Extraordinary results, quality of work or extremely well structured and justified report. There might still be some very minor possibilities for improvement.
 - Repository is clear and runnable. Report is well organized, results are discussed, visualizations are added value to the text and well prepared. Apart from minimum criteria the group tried multiple of their novel ideas to approach the problem.
- 9 - very good: Above average knowledge presentation with some errors.
 - Same as above - it can be visible that a group had novel ideas but they got out of time or did not finish (polish) everything. The submission has multiple minor flaws.
- 8 - good: Submission of solid work, mainly addressing given instructions only.
 - Group implemented everything suggested by the minimum criteria but not investigated further (not found much related work, did not apply multiple other techniques, ...).
- 7 - superficial: Below average knowledge and submission of work with errors that show partial understanding.

Group implemented everything suggested by the minimum criteria but did not discuss results well, performed simple analyses only, ... Also the report is not well organized, and lacks data for reproducibility.

- 6 - sufficient: Minimum criteria addressed with some major errors and drawbacks.

Group was trying to implement minimum criteria (or part only) but their work has many minor flaws or a few major ones. The report also reflects their motivation.

- 5 - insufficient: Too much lack of knowledge, too many major errors or no work-effort could be drawn from the submitted work.

The group did not address one or more points of the minimum criteria and the report contains major flaws. It can be seen that the group did not invest enough time into the work.

Final project preparation guidelines and peer review instructions

Some major remarks that you should keep in mind are the following regarding final submission:

- Comment on all the specifics of your algorithms that you use or have designed (i.e. features, hyperparameters, ...). In some cases it is useful to include an image instead of providing long descriptions.
- When you include graphs/images, they must be readable and provide additional insight (e.g. if there are lots of lines across the image or little space between them, it is not okay). When you report on results, keep them as much as possible in one table, so that a reader can compare different configurations. Also, it is useful to bold the best results.
- Both images and tables should be self-contained - i.e., together with the caption they need to provide enough information to the reader to understand it's meaning without reading text around.
- Keep your report concise and try to not submit a report longer than 4 pages + references + appendices. Also, make sure you follow the proposed template.
- Focus on reporting results and using sensible measures. Try to find examples where your algorithm works better and may not even work at all. Explain why and also justify the differences in approaches that you used. In case previous work exists for your dataset, put the best results of other researchers in your results table (even if your results are much lower).
- Your submitted work (repository and report) should be structured in a way that your colleagues would be able to understand and re-run everything. Include all dependencies for your projects:
 - In case you have used a non-public (or semi-public) dataset, do not include it in the repository, just put your contact data or protected link to download data/provide other instructions to retrieve data.
 - Datasets that are available elsewhere should be just linked in your report/repository. If you performed additional transformations on datasets, scripts for that should be available in the repository.
 - Some of you used models that take longer to train. You can include those models (maybe just the best one) in the repository or elsewhere and link it.
- Lastly, check that your repositories are publicly available before the peer review period starts! **IMPORTANT:**
 - Include links to all dependencies/corpora or include them in the repository, so that anyone can check your work. Also include annotated data (if you manually prepared a corpus) or trained models (if training takes a long time).
 - For anyone that will review your work, it must be as simple as possible to run your code.

Peer review instructions:

1. Please find the projects you need to review (see link above).
2. Each group needs to review projects of the same topic they have chosen.
3. Submit your peer review scores in the Google Form (see link above).
4. You will get a score also for your grading, depending on how much (of course by some margin) your grading will be different from the assistant's grading.
5. Follow the scoring criteria as presented above and include feedback to your mark.

Final project presentation instructions

Each group will have **max. 3 minutes (STRICT)** to present their project. I will put your report to the projector and you will present it along with your report. I propose that you focus on specific interesting part that you will present (e.g., table, graph, figure, ...). The most important aspect to present is:

What is the "**take-away message**" of your work? This should be concrete and concise, so that anyone can understand (also a completely lay person).

See timetable above for time slots of your presentation. If you cannot attend, please write to me to get an alternative time slot.

Project: Domain-Specific AI Assistant

Description: Students will develop a specialized conversational AI assistant tailored to a specific institutional domain, technical & IT support, legal advising, etc. While general-purpose Large Language Models (LLMs) are highly capable, they often lack the localized context required to answer specific, institution-dependent questions accurately and without hallucination. This project focuses on adapting an LLM to serve as a reliable, domain-specific expert. The project will involve defining a strict use case, preparing a knowledge bases and tools (e.g., extracting data from university websites, formatting PDF rulebooks), and implementing techniques like Retrieval-Augmented Generation (RAG) or parameter-efficient fine-tuning (PEFT) to inject contextual knowledge. The final system should effectively interpret user intent, retrieve the correct information, process it, and deliver accurate answers.

Proposed methodology:

- **Literature review:** Conduct a survey of current approaches for injecting domain-specific knowledge into LLMs. Compare the trade-offs between advanced prompt engineering, RAG, and fine-tuning for specialized applications, particularly in educational or administrative contexts.
- **Data curation and domain definition:** Define the specific domain (e.g., "Faculty Student Rulebook and FAQs"). Acquire, clean, and format relevant datasets, such as official university regulations, course syllabi, and administrative guidelines.

- **Implementation:** Build the assistant pipeline. If utilizing RAG, implement document chunking, vector embeddings, and retrieval mechanisms. Experiment with system prompt engineering (e.g., defining a persona, setting guardrails to prevent out-of-domain answers) to ensure the assistant remains focused, polite, and factually accurate.
- **Evaluation:** Assess the model's performance using both automated and human-centric metrics. Utilize metrics for retrieval accuracy and relevance, and conduct human evaluations focusing on factual consistency, tone, and the system's ability to safely handle ambiguous or unanswerable queries (e.g., knowing when to direct the student to a human advisor).
- **Final report:** Prepare a comprehensive report detailing the architecture, the challenges of adapting general models to specific domains, and insights into how such AI assistants can streamline administrative workflows in real-world educational settings.

Proposed Datasets

Students are required to select or curate a knowledge base that serves as the "source of truth" for their assistant. The project should utilize one of the following sources:

- Automatic answering end user and public consultation questions for the *Agency for Communication Networks and Services of the Republic of Slovenia* (<https://www.akos-rs.si>). Your goal will be retrieve AKOS' data, organize it, and use it for the modelling part and models building.
- *Student affairs office at UL FRI* – around 2000 enrolled students at our faculty are potential customers of our student office. Some of the questions could be easily answered from the online sources. Your goal is to gather and organize Web-accessible data (e.g., UL FRI web page, UL documents, polls with your colleagues; see example chatbot at UL FS), prepare models and evaluate them.
- *Law domain* – within the LLM4DH project we have prepared a large law corpus containing Slovenian legislation. Your goal would be to process the data, think of law-related use cases (e.g., contract preparation help), prepare models and evaluate them. Some examples of general use cases are <https://www.mojmalipravnik.net>, or <https://app.pravko.si>.
- *Your Custom Project* – for the custom project, you need to prepare a proposal and get confirmation by the assistant during the lab sessions. Within the proposal especially address and describe (a) possible data sources and (b) outline use cases your model would support.

Later, we might provide you additional rules for packaging your work, so that peer-review and reproducibility of your work will be easier.

References:

- Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS.
<https://arxiv.org/abs/2005.11401>
- Hu, E. et al. (2021). *LoRA: Low-Rank Adaptation of Large Language Models*.
<https://arxiv.org/abs/2106.09685>
- Schick, T. et al. (2023). *Toolformer: Language Models Can Teach Themselves to Use Tools*.
<https://arxiv.org/abs/2302.04761>
- Song, Z. et al. (2025). *Injecting Domain-Specific Knowledge into Large Language Models: A Comprehensive Survey*, arXiv.org. Available at: <https://arxiv.org/abs/2502.10708>

Project 2 (**only for Digital Linguistics students!**): Translation Assistant with Retrieval-Augmented Generation (RAG)

Description:

Students will build a translation assistant that helps users translate documents from a chosen domain with highly specialized terminology (e.g., study rules, course descriptions, administrative procedures) more accurately and consistently by grounding translations in an approved set of reference materials. General-purpose translation tools and LLMs often produce fluent output but may miss institutional terminology, produce inconsistent wording, or hallucinate. This project addresses that by using Retrieval-Augmented Generation (RAG): the assistant first retrieves relevant passages from curated institutional sources (web pages, PDFs, glossaries, style guides) and then uses them to produce terminology-consistent, source-grounded translations with short citations to the retrieved text.

Proposed methodology:

1. Domain definition and data preparation: Define a specific translation domain. Collect and organize relevant resources (documents, glossary, ...). Clean, structure, and prepare the corpus for retrieval.
2. Preprocessing: Convert documents into consistent textual format and segment them into retrieval-ready units while preserving source traceability.
3. Implementation (Retrieval-Augmented Translation): Index the corpus in a lightweight vector retrieval system. For each translation query, retrieve relevant passages and generate a translation grounded in retrieved content and institutional terminology.
4. Grounded output: Return some relevant information: the translation, a short terminology justification, citations to the retrieved source passages, ...

5. Evaluation: Assess various aspects, e.g., terminology consistency, factual alignment with sources, and overall translation quality. Compare grounded and non-grounded outputs.
6. Final report: Describe the corpus, pipeline design, grounding strategy, evaluation results, and practical implications for institutional translation tasks.

Resources: [Instructions](#) on how to access GPU resources.

References:

- Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS.
<https://arxiv.org/abs/2005.11401>